



C.5 Doppler effect

Wave behaviour · IB Diploma · Concept guide and original exam-style practice

Understand the concept

The Doppler effect is the change in observed frequency (or wavelength) of a wave due to relative motion between source and observer: frequency increases when source and observer approach each other and decreases when they recede. For sound waves (source moving, observer stationary, speed of sound v), the observed frequency is $f_{\text{obs}} = f_{\text{source}} \cdot v / (v \pm u_s)$, with the sign depending on whether the source approaches ($-$, giving higher f_{obs}) or recedes ($+$, lower f_{obs}); analogous expressions exist for a moving observer. For light, at speeds much less than c the fractional shift in wavelength/frequency approximates $\Delta f/f \approx \Delta\lambda/\lambda \approx v/c$, used in astronomy to measure the redshift (receding) or blueshift (approaching) of distant sources such as galaxies, revealing radial velocity.

Worked example: follow the reasoning

A police car siren emits sound at 600 Hz and moves towards a stationary observer at $30 \text{ m}\cdot\text{s}^{-1}$. Speed of sound in air is $340 \text{ m}\cdot\text{s}^{-1}$. Find the observed frequency.
 $f_{\text{obs}} = f_{\text{source}} \times v / (v - u_s) = 600 \times 340 / (340 - 30) = 600 \times 340 / 310 = 600 \times 1.097 = 658 \text{ Hz}$.
Answer: $f_{\text{obs}} \approx 658 \text{ Hz}$.

Quick check · multiple choice

A star's light is observed to have its spectral lines shifted to longer wavelengths (redshifted) compared to laboratory values. What does this indicate about the star's motion relative to Earth?

- A Moving towards Earth
- B Moving away from Earth
- C Stationary relative to Earth
- D Moving perpendicular to the line of sight only

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 An ambulance travels at a constant speed towards a stationary observer, sounding its siren at a source frequency of 500 Hz. The observer hears a frequency of 560 Hz. Take the speed of sound in air as $340 \text{ m}\cdot\text{s}^{-1}$. [7]
- (a) State whether the ambulance is moving towards or away from the observer, with a reason.
- (b) Calculate the speed of the ambulance.
- (c) Calculate the frequency the observer would hear immediately after the ambulance passes and moves away at the same speed.

Final answer: _____



Answers and explanation

Question 1 [7]

(a) Towards the observer, because the observed frequency (560 Hz) is higher than the source frequency (500 Hz), consistent with an approaching source (compressed wavefronts).

(b) $f_{\text{obs}} = f_{\text{source}} \cdot v / (v - u_s) \rightarrow 560 = 500(340) / (340 - u_s) \rightarrow 340 - u_s = 500 \times 340 / 560 = 303.6$
 $\rightarrow u_s = 340 - 303.6 = 36.4 \text{ m} \cdot \text{s}^{-1}$.

(c) Receding: $f_{\text{obs}} = f_{\text{source}} \cdot v / (v + u_s) = 500(340) / (340 + 36.4) = 170000 / 376.4 = 451.6 \text{ Hz} \approx 452 \text{ Hz}$.

Quick check answer

B. A redshift (increased wavelength) indicates the source is receding from the observer, consistent with the Doppler effect for light.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Physics HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.